

Dajen Torralba

Software Engineer • Systems Development • AI • AWS Cloud

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SUMMARY

A software engineer with an extensive professional experience in designing, developing, and deploying production-level software, intelligent systems, cybersecurity modules, and comprehensive ML/AI models. Holds AWS Certified Cloud Practitioner and AWS Certified AI Practitioner certifications, supporting a strong foundation in AI and modern distributed systems. Possesses a solid academic foundation in computer science (BS), enhanced by practical expertise in C, C++, and Python development, agile-focused engineering practices, and improving efficiency and workloads with AI. Demonstrates a track record of adaptability, efficiency, honor, and excellence – dynamically driving results-oriented solutions for the benefit of a technologically-advanced society.

SKILLSET

Programming and Software Development

- C, C++, Python
- APIs, systems programming
- Unit & Integration Testing (PyTest, GTest)

Cloud and AWS

- AWS Cloud Practitioner concepts
- AWS AI Practitioner concepts
- Cloud architecture fundamentals & distributed computing

DevOps, CI/CD & Automation

- Docker, Bamboo pipelines, automated deployments
- Git: pull requests, branching, merge conflict resolution

AI, ML & Intelligent Systems

- Claude, AWS AI models & tooling, local LLM deployment
- ML model training, development, & evaluation (classification, regression)
- Prompt engineering, model evaluation, workflow automation
- Robotics & intelligent systems (ROS, ML model integration)
- Data Processing and Analytics

Tools & Development Environments

- VS Code, Visual Studio, JetBrains
- Jira, Confluence, Bitbucket/Stash, Docker

Leadership, Delivery, and Software Engineering Practices

- Scrum Master experience & Agile/SAFe delivery

EXPERIENCE

Engine Software Engineer

Aug 2025 - Mar 2026

Trend AI / Trend Micro Inc.

- Developed and maintained high-performance engine and kernel modules supporting threat detection and real-time system analysis.
- Applied Windows Internals expertise and API hooking techniques to build reliable, low-latency system components.
- Contributed to AI-driven automation initiatives to improve code delivery, testing, and operational efficiency.
- Collaborated with cross-functional teams to support contingency planning, rapid response, and system reliability efforts.
- Participated in code reviews and engineering design discussions to ensure maintainability and performance.

Robotics Software Engineer

Sep 2023 - Aug 2025

Dyson Electronics Pte. Ltd.

- Built and optimized robotics functions including particle detection, classification, feature matching, navigation, and localization.
- Designed and maintained inference pipelines for real-time sensor processing and decision-making.
- Served as Team Lead for high-complexity epics and acted as Scrum Master, facilitating planning, ceremonies, and cross-team coordination.
- Developed unit and integration tests to ensure reliability, coverage, and regression stability across robotics modules.
- Participated in code reviews and collaborated with hardware, firmware, and algorithm teams to deliver integrated solutions.

Application Developer Intern

Jun - Aug 2022

Texas Instruments Inc.

- Developed a web-based data retrieval feature used across test and business functions.
- Improved departmental efficiency and productivity by an estimated 60–70% through the delivered solution.

IT Helpdesk Intern

Jan - Mar 2019

Al Jazirah Vehicles Agencies Co. Ltd.

- Provided IT support focused on user management, system access, and troubleshooting.

PROJECTS

Real-Time Property Demand Heatmap

- Developed a full-stack web platform that aggregates live data from property listings, YouTube, Reddit, and Google Trends to visualize neighborhood demand across Riyadh on an interactive heatmap.
- Built a Python FastAPI backend with PostgreSQL, a configurable composite scoring model with admin-adjustable signal weights, JWT authentication, and an audit-logged admin dashboard.
- Containerized the application using Docker Compose and deployed a Next.js/TypeScript frontend.

Instagram FollowCheck

- Built a privacy-first web tool for analyzing Instagram follower relationships from exported account data.
- Engineered a client-side JSON parser in TypeScript that processes Instagram data exports entirely in the browser — no data is sent to any server.
- Deployed as a cross-platform application supporting web (Vite/React), Docker, and native desktop builds (Electron for macOS and Windows).

Trend Vision One™ AI Security

- Supported development of AI-enhanced security components integrated into enterprise-grade detection systems.

Dyson 360 Vis Nav™ – Intelligent Machines

- Developed features and committed bug fixes under perception, mapping, and navigation domains for Dyson's IM.

Prediction of Human Central Features using a PLP-CNN Voice Input Approach (Thesis)

- Developed an end-to-end ML system capable of predicting age, sex, height, and weight from voice input.
- Built and trained deep learning models using PyTorch and TensorFlow, including preprocessing, feature extraction, and evaluation.
- Deployed the model through a lightweight web application enabling voice upload and real-time inference.

A Computer Vision Deep Learning Classifier for Obscene and Inappropriate Media Content

- Designed and trained a CNN-based classifier for image content moderation.

LICENSES AND CERTIFICATIONS

AWS Certified AI Practitioner (AIF-C01)

- Issued by Amazon Web Services
- Licensed from May 2026 - May 2029
- Validation ID: 560c01eee0bb447b8ff0b57cf965ce9

AWS Certified Cloud Practitioner (CLF-C02)

- Issued by Amazon Web Services
- Licensed from May 2026 - May 2029
- Validation ID: dd1319a0852e492588a6de2f751a08e7

LEADERSHIP & EXTRACURRICULARS

Epic Lead & Scrum Master, Dyson Electronics Pte. Ltd.

- Led high-complexity engineering epics, facilitated Scrum ceremonies, coordinated cross-team planning, and supported delivery of robotics intelligence features.

Graduate, Dyson Electronics Pte. Ltd.

- Participated on career-development programs.
- Shared insights on engineering development, leadership, and transitioning into technical roles.

Project Leader, Saint Louis University (2019–2023)

- Directed project execution, research initiatives, and team coordination.

Secretary-General, Saint Louis University Congress of Louisians (2022–2023)

- Oversaw documentation, communication, and organizational governance for the university's central student body.

Budget & Appropriations Scribe, Saint Louis University Congress of Louisians (2021–2022)

Vice-President, Saint Louis University Integrated Confederacy (2021–2022)

Chairperson, RIGHTS Saint Louis University (2021–2022)

Director for Communications, Saint Louis University Executive Central Committee (2020–2021)

Director for Media Relations, RIGHTS Saint Louis University (2020–2021)

Secretary, Saint Louis University SAMCIS School Assembly (2019–2020)

EDUCATION

Bachelor of Science in Computer Science, Saint Louis University

Baguio City, Philippines

- Graduated **Cum Laude** with a GWA of 90.73.
- **JLSS Scholar** under the Department of Science and Technology (2021–2023).
- **Dean's Lister** for all academic years (2019–2023).
- Completed advanced coursework in Machine Learning, Data Mining, Algorithms, Operating Systems, Computer Vision, and Software Engineering.

Diploma in STEM, Future Generation Philippine International School

Riyadh City, Kingdom of Saudi Arabia

- Graduated **With Honors** with a GWA of 93.268
- **Model Student of the Year** Awardee alongside other honorable mentions.